

Proposal - Real Estate Agency X (Image Classification)



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Introduction

Real Estate Agency X receives approximately 50,000 new properties for sale across Australia on a monthly basis. Consequently, the image classification for these properties on listing websites is vastly time consuming, as it requires manual review and upload by staff members.

To upload individual property photos into predefined categories (e.g. living room, bedroom, kitchen), the process takes 20 min on average for each property to be manually uploaded. Especially with 50,000 additional properties every month, this approach is highly inefficient.

This proposal recommends the deployment of an image classification model to automate the process. To ensure effectiveness and efficiency, we will apply the CRISP-ML framework to outline the project plan.

A. Business Problem and Impact

Before we design, develop, and deploy the product, it is critical to highlight the underlying business problem and its impact to performance. The current manual approach can lead to the following issues:

1. Decreased Employee Satisfaction - staff frequently complained about the repetitive upload process, which encourages frustration, lowers motivation, and increases burnout risk. Such monotonous tasks are well documented to reduce motivation and increase turnover likelihood, while also harming productivity and health (Rahman & Sen, 1987).
2. Inefficient Resource Allocation - as each property require about 20 min to upload into predefined categories, the additional 50,000 properties each month could cost Agency X around 16,650 hrs ($20 * 50,000$) of work per month just to classify and upload images. This restrains capacity for more meaningful tasks like market research and client support to build value.
3. Limitation of Scalability - due to the resource intensity of this process, property intake is also limited to the number of staff members available. Labour cost will increase proportionally to increased property influx, requiring additional salaries and training, which hinders the Agency's ability to expand.

B. ML-Based Solution and Requirements

1. Solution

Deploying a ML classification model can effectively address the identified issues that are undermining performance through automated image classification. This would remove the monotonous work performed by staff to prevent burnouts and frustration to engage in higher value tasks to improve satisfaction.

As the system can process photos in seconds compared to manual processing, staff and resources can contribute to other productive activities that support growth and reduce operational costs. Furthermore, the model also enables scalability through processing large amounts of properties beyond 50,000 with greater accuracy and speed without increased labour and training. This will enable Agency X to gain competitive advantage through cost reduction and efficient resource reallocation by eliminating the additional 16,650 hrs of work using ML.

2. Requirements

Technical Feasibility

Requirements	Details
Scalability	The system will need to handle increased data volume if property intake increases, and account for increasing number of photos for classification.
Integration	The system should also be able to integrate into existing systems and processes. The need for additional tools or systems should be evaluated.
Maintenance	The system should be maintained after deployment by monitoring photo classification metrics and fixing errors when occurred.

Data Quality

Data Relevance	Ensure data for training and validating is related to the predefined property categories (e.g. bedroom, kitchen).
Data Consistency	Check for duplicate photos and ineligible image formats.
Data Bias	Data used should include a diverse range of property types and conditions to avoid bias in the model.

Model Performance

Accuracy	The model should deliver accurate image classification across diverse conditions, including urban and suburban environments.
Generalisation	Validate and test that the model can perform under new data to reduce risks of overfitting or underfitting. Use a diverse dataset by ensuring photos from each category have a fair distribution.
Latency	The system should execute photo categorisation at high speed, and continue to perform without latency problems even with exponential growth of properties.

Ethics & Legalities

Transparency	Stakeholders should be informed about the usage of AI in the system. The purpose and context should be reasonably disclosed for transparency (Department of Industry, 2019).
Privacy	Images used for model training, validation, and deployment should only be employed for the disclosed purpose. Published photos should not include sensitive information of clients.
Bias	To reduce misrepresentation and misclassification, listings should feature a diverse mix of properties such as rural and old-fashioned homes, rather than focusing predominantly on modern apartments and new urban properties.

C. Project Proposal (CRISP-ML)

The following pages will provide a comprehensive framework and plan of the project to outline the model lifecycle using a structured approach. The CRISP-ML lifecycle (Cross-Industry Standard Process for the development of Machine Learning applications with Quality assurance methodology) consists of six phases that will be addressed.

1. Business and Data Understanding

Business Objectives: To automate photo classification of properties and upload into predefined categories. The project seeks to fulfill three objectives:

1. Improve Employee Satisfaction
2. Reduce Operational Costs
3. Reallocate Resources for Growth

Business Success Criteria: KPIs such as 360-Degree Feedback Score can measure if staff productivity and satisfaction improved. While property intake can act as an indicator of business growth, where the model should handle photos from above 70,000 properties. In addition, comparing labour costs before and after deployment can prove cost savings.

ML Success Criteria: F1-Score is recommended to measure model performance. The dataset contains images from diverse categories, where F1-Score is useful in accounting for imbalanced datasets. A score above 0.7 generally shows a good balance between precision and recall (Sentner, n.d).

Technical Feasibility: Only modest hardware equipped with a modern GPU and decent CPU is required for computational power. Can be scaled with additional GPU machines.

Legal Compliance: Ensure the system is compliant with the Privacy Act 1988 and follows Australia's AI Ethics Principles (Department of Industry, 2019).

Data Understanding: Image formats should be consistent (e.g. PNG, JPEG) and photo size should be standardized and quality should be clear for data quality. Dataset attributes should include category (e.g. kitchen, bathroom, garage), image file, property ID, property type, and geographic information.

2. Data Engineering

Data Selection: Images should have a fair distribution from each category and property types to avoid biases. Training dataset should include at least 10,000 labeled images and scaled if necessary.

Data Cleaning: Remove missing, inconsistent, duplicates, and outliers from the dataset. This includes duplicate photo files, missing data, unclear images, or images not included in the class labels.

Data Construction: Utilize data augmentation to construct new data by applying transformation to images, which creates additional samples to improve generalization.

Data Normalisation: For image classification tasks, normalisation is often not required. But if necessary, normalise RGB values to a standardised range.

3. ML Model Engineering

Model Selection: Based on the problem statement, data, and business objectives, the following models can be employed:

- YoloV5 (Object Detection) - employed in a similar case study for global auto-tagging of real estate images (Florea et al., 2022).
- VGG16 (CNN) - highlighted by paper that it achieved the highest accuracy on test data for classification of real estate images (Nejad et al., 2025).
- Random Forest - most simple approach for photo classification with multiple classes but risks accuracy in comparison to Deep Learning models.
- Multinomial Logistic Regression - another simple multi-class image classification.

Model Training: Generally the model will be trained on 80% of the dataset, while the rest will be employed for validation and testing.

Model Registration: Model performance will be tracked through documentation and meta-data can be stored and analysed using tools such as MLflow. This manages version control, storing, and tracking of each model to facilitate collaboration.

4. ML Model Evaluation

Model evaluation is a crucial phase in the development lifecycle which determines if the model meets the defined success criteria. Besides meeting KPI targets, as suggested above, F1-Score is an appropriate evaluation metric for multi-class classification with an imbalanced dataset. Thereby, F1-Score will be prioritized ($\geq 70\%$) as it considers both precision and recall. If evaluation value is unsatisfactory, iterate model training through adjusting hyperparameters to optimize the model. In addition, other soft measures such as robustness, explainability, and latency should also be considered.

5. Model Deployment

The Canary deployment strategy will be used for deployment to reduce risks of error when introducing a new system. This will be achieved through gradual implementation into operations by testing the model on properties in incremental amounts (e.g. 10% automation 90% manual) until performance and usability is acceptable for full deployment.

6. Model Monitoring and Maintenance

Without continued monitoring and maintenance, model performance can degrade which results in false predictions due to ML drift and changes in the dynamics of the environment. For that reason, input signals should be monitored by comparing the statistics of incoming data to training data statistics. If monitoring results display signs of performance degradation, model update should be considered using new property photos to re-train the model (fine-tune).

D. MLOps Proposal for Development

The extent of automation in the ML process depends on the maturity levels of MLOps (level 0 - 2) which are principles and practices in ML system automation.

In the case of Real Estate Agency X, the organisation is recommended to adopt MLOps level 0 (manual process) as a starting point, and shift to level 1 once operation expands and the production environment adapts to ML application.

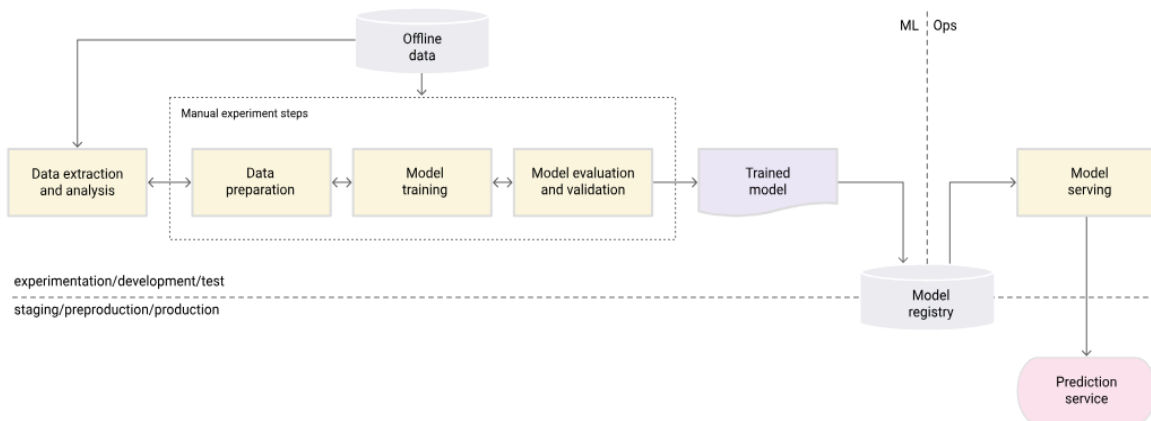


Figure 1. Manual ML steps in level 0 MLOps (Google Cloud, 2024)

MLOps Suggestion:

- Adopt level 0 as a manageable starting point, as the agency is only starting to apply ML into operations. This requires manual processing and active monitoring of the ML system.
- Transition to level 1 at a later stage as trends and property styles change over time in photos, which will require frequent model monitoring and updating to avoid degradation of performance.
- This will establish automated training pipelines that will continuously retrain the model based on new property images as prediction service.

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